Hierarchical Clustering: Average Link and Complete Link

Given Distance Matrix (Table 1)

$$\begin{bmatrix} 0.00 & 0.10 & 0.41 & 0.55 & 0.35 \\ 0.10 & 0.00 & 0.64 & 0.47 & 0.98 \\ 0.41 & 0.64 & 0.00 & 0.44 & 0.85 \\ 0.55 & 0.47 & 0.44 & 0.00 & 0.76 \\ 0.35 & 0.98 & 0.85 & 0.76 & 0.00 \end{bmatrix}$$

Points: P_1, P_2, P_3, P_4, P_5

1. Average Linkage (UPGMA)

For Average Link, the distance between two clusters C_i and C_j is:

$$d(C_i, C_j) = \frac{1}{|C_i| \cdot |C_j|} \sum_{p \in C_i} \sum_{q \in C_j} d(p, q)$$

Step 1: Initial clusters

All points are singleton clusters:

$$\{P_1\}, \{P_2\}, \{P_3\}, \{P_4\}, \{P_5\}$$

Step 2: Find closest pair

From the distance matrix, the minimum distance is 0.10 between P_1 and P_2:

$$\{P_1, P_2\}$$

Step 3: Update distances to new cluster

$$d(\{P_1, P_2\}, P_3) = \frac{d(P_1, P_3) + d(P_2, P_3)}{2} = \frac{0.41 + 0.64}{2} = 0.525$$
$$d(\{P_1, P_2\}, P_4) = \frac{d(P_1, P_4) + d(P_2, P_4)}{2} = \frac{0.55 + 0.47}{2} = 0.51$$
$$d(\{P_1, P_2\}, P_5) = \frac{d(P_1, P_5) + d(P_2, P_5)}{2} = \frac{0.35 + 0.98}{2} = 0.665$$

Updated distance matrix:

$$\begin{bmatrix} 0 & 0.525 & 0.51 & 0.665 \\ 0.525 & 0 & 0.44 & 0.85 \\ 0.51 & 0.44 & 0 & 0.76 \\ 0.665 & 0.85 & 0.76 & 0 \end{bmatrix}$$

Step 4: Merge closest clusters

Closest distance is 0.44 between P_3 and P_4:

$$\{P_3, P_4\}$$

Step 5: Update distances

$$d(\{P_1, P_2\}, \{P_3, P_4\}) = \frac{0.41 + 0.64 + 0.55 + 0.47}{4} = 0.5175$$
$$d(\{P_3, P_4\}, P_5) = \frac{0.85 + 0.76}{2} = 0.805$$
$$d(\{P_1, P_2\}, P_5) = 0.665 \text{ (from previous step)}$$

Updated clusters:

$$\{P_1, P_2\}, \{P_3, P_4\}, \{P_5\}$$

Step 6: Merge closest clusters

Distance matrix:

$$\begin{bmatrix} 0 & 0.5175 & 0.665 \\ 0.5175 & 0 & 0.805 \\ 0.665 & 0.805 & 0 \end{bmatrix}$$

Closest distance is 0.5175:

$$\{P_1, P_2, P_3, P_4\}, \{P_5\}$$

Step 7: Merge remaining clusters

Distance:

$$d(\{P_1, P_2, P_3, P_4\}, \{P_5\}) = \frac{0.35 + 0.98 + 0.85 + 0.76 + 0.55 + 0.47 + 0.44 + 0.44}{8} = 0.57$$

Final cluster:

$$\{P_1, P_2, P_3, P_4, P_5\}$$

Dendrogram: Average Link

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=== Average Linkage ===
Step-by-step clustering:
Step 1: Merge ['P1'] and ['P2'] -> ['P1', 'P2'], Distance = 0.100
Step 2: Merge ['P3'] and ['P4'] -> ['P3', 'P4'], Distance = 0.440
Step 3: Merge ['P1', 'P2'] and ['P3', 'P4'] -> ['P1', 'P2', 'P3', 'P4'], Distance = 0.518
Step 4: Merge ['P5'] and ['P1', 'P2', 'P3', 'P4'] -> ['P5', 'P1', 'P2', 'P3', 'P4'], Distance = 0.735

Final cluster: ['P5', 'P1', 'P2', 'P3', 'P4']
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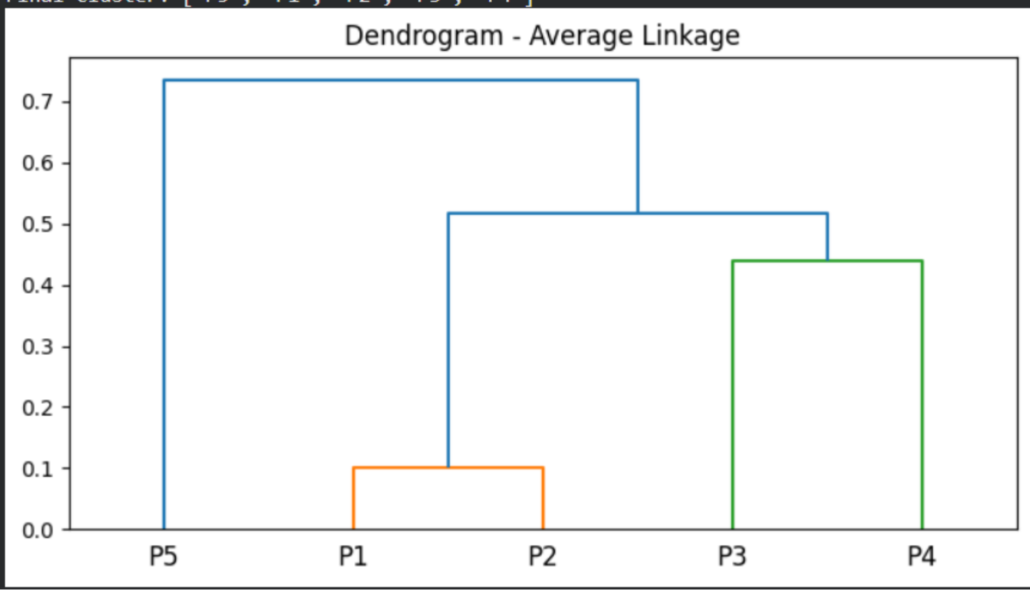

Figure 1: Dendrogram - Average Linkage

2. Complete Linkage (MAX)

For Complete Link, the distance between two clusters C_i and C_j is:

$$d(C_i, C_j) = \max_{p \in C_i, q \in C_j} d(p, q)$$

Step 1: Initial clusters

All points are singleton clusters:

$$\{P_1\}, \{P_2\}, \{P_3\}, \{P_4\}, \{P_5\}$$

Step 2: Merge closest pair

Minimum distance is 0.10 between P_1 and P_2:

$$\{P_1, P_2\}$$

Step 3: Update distances to new cluster

$$d(\{P_1, P_2\}, P_3) = \max(d(P_1, P_3), d(P_2, P_3)) = \max(0.41, 0.64) = 0.64$$
$$d(\{P_1, P_2\}, P_4) = \max(0.55, 0.47) = 0.55$$
$$d(\{P_1, P_2\}, P_5) = \max(0.35, 0.98) = 0.98$$

Step 4: Merge closest pair

Next closest is P_3 and P_4 at 0.44:

$$\{P_3, P_4\}$$

Step 5: Update distances

$$d(\{P_1, P_2\}, \{P_3, P_4\}) = \max(0.64, 0.55) = 0.64$$
$$d(\{P_3, P_4\}, P_5) = \max(0.85, 0.76) = 0.85$$

Step 6: Merge next closest clusters

Closest is $d(\{P_1, P_2\}, \{P_3, P_4\}) = 0.64$:

$$\{P_1, P_2, P_3, P_4\}, \{P_5\}$$

Step 7: Merge remaining clusters

$$d(\{P_1, P_2, P_3, P_4\}, P_5) = \max(0.98, 0.85) = 0.98$$

Final cluster:

$$\{P_1, P_2, P_3, P_4, P_5\}$$

Dendrogram: Complete Link

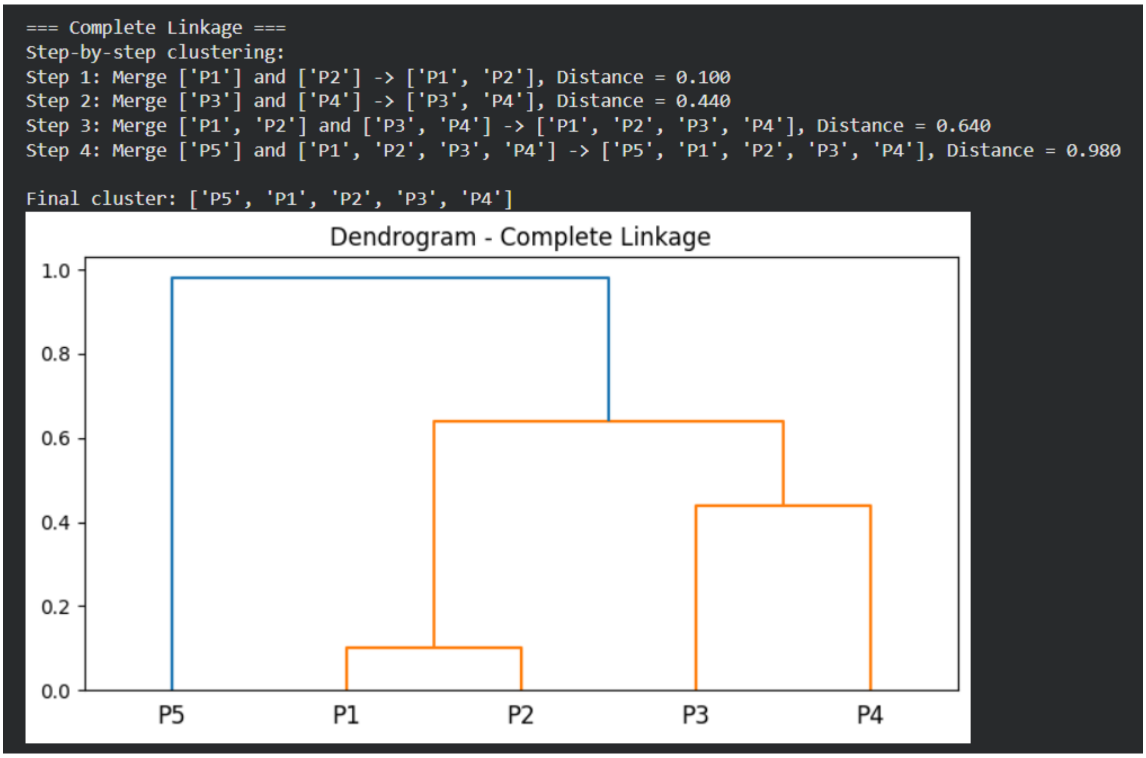

Figure 2: Dendrogram - Complete Linkage

3. Single Linkage (MIN)

For Single Link, the distance between two clusters C_i and C_j is:

$$d(C_i, C_j) = \min_{p \in C_i, q \in C_j} d(p, q)$$

Step 1: Initial clusters

$$\{P_1\}, \{P_2\}, \{P_3\}, \{P_4\}, \{P_5\}$$

Step 2: Merge closest pair

Minimum distance is 0.10 between P_1 and P_2:

$$\{P_1, P_2\}$$

Step 3: Update distances to new cluster

$$d(\{P_1, P_2\}, P_3) = \min(0.41, 0.64) = 0.41$$
$$d(\{P_1, P_2\}, P_4) = \min(0.55, 0.47) = 0.47$$
$$d(\{P_1, P_2\}, P_5) = \min(0.35, 0.98) = 0.35$$

Step 4: Merge next closest cluster

Distance 0.35 between $\{P_1, P_2\}$ and P_5:

$$\{P_1, P_2, P_5\}$$

Step 5: Update distances

$$d(\{P_1, P_2, P_5\}, P_3) = \min(0.41, 0.64, 0.85) = 0.41$$
$$d(\{P_1, P_2, P_5\}, P_4) = \min(0.47, 0.76, 0.76) = 0.47$$

Step 6: Merge next closest cluster

Distance 0.41 between $\{P_1, P_2, P_5\}$ and P_3:

$$\{P_1, P_2, P_3, P_5\}, \{P_4\}$$

Step 7: Merge remaining clusters

Distance 0.44 between $\{P_1, P_2, P_3, P_5\}$ and P_4:

$$\{P_1, P_2, P_3, P_4, P_5\}$$

Single Linkage Dendrogram

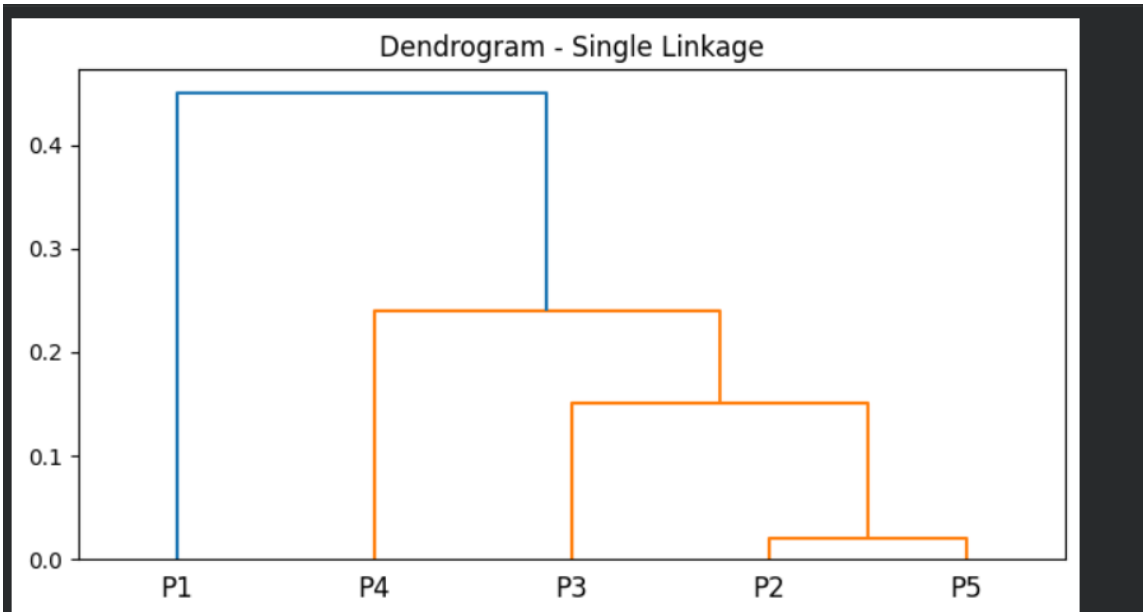

Figure 3: Dendrogram - Single Linkage

Given Similarity Matrix (Table 3)

$$\begin{bmatrix} 1.00 & 0.10 & 0.41 & 0.55 & 0.35 \\ 0.10 & 1.00 & 0.64 & 0.47 & 0.98 \\ 0.41 & 0.64 & 1.00 & 0.44 & 0.85 \\ 0.55 & 0.47 & 0.44 & 1.00 & 0.76 \\ 0.35 & 0.98 & 0.85 & 0.76 & 1.00 \end{bmatrix}$$

Points: P_1, P_2, P_3, P_4, P_5

Step 1: Convert Similarity to Distance

Distance is computed as:

$$D = 1 - S$$

$$\begin{bmatrix} 0.00 & 0.90 & 0.59 & 0.45 & 0.65 \\ 0.90 & 0.00 & 0.36 & 0.53 & 0.02 \\ 0.59 & 0.36 & 0.00 & 0.56 & 0.15 \\ 0.45 & 0.53 & 0.56 & 0.00 & 0.24 \\ 0.65 & 0.02 & 0.15 & 0.24 & 0.00 \end{bmatrix}$$

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1. Single Linkage Clustering

Rule: distance between clusters = minimum distance between points.

Step-by-step merging:

1. Closest pair: P_2 and P_5 (distance = 0.02) $\Rightarrow$ merge to $\{P_2, P_5\}$
2. Update distances:

   $$d(\{P_2, P_5\}, P_1) = 0.65, \quad d(\{P_2, P_5\}, P_3) = 0.15, \quad d(\{P_2, P_5\}, P_4) = 0.24$$

3. Next closest: $\{P_2, P_5\}$ and P_3 (0.15) $\Rightarrow$ merge $\{P_2, P_3, P_5\}$
4. Update distances:

   $$d(\{P_2, P_3, P_5\}, P_1) = 0.59, \quad d(\{P_2, P_3, P_5\}, P_4) = 0.24$$

5. Next closest: $\{P_2, P_3, P_5\}$ and P_4 (0.24) $\Rightarrow$ merge $\{P_2, P_3, P_4, P_5\}$
6. Final merge: P_1 and $\{P_2, P_3, P_4, P_5\}$ (0.45) $\Rightarrow$ cluster $\{P_1, P_2, P_3, P_4, P_5\}$

Dendrogram: Single Linkage

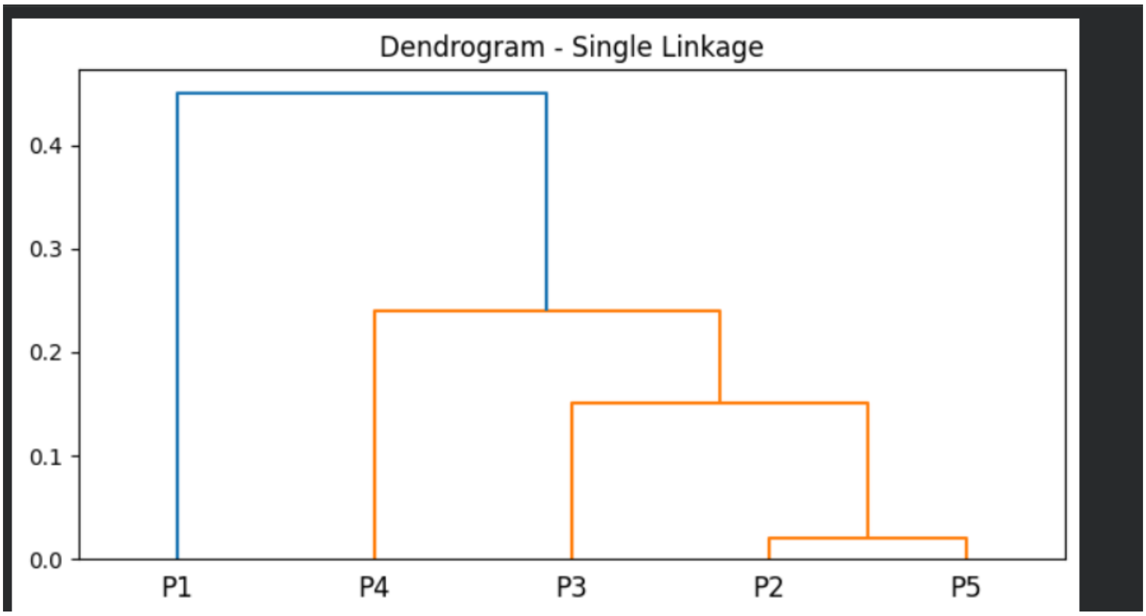

Figure 4: Dendrogram for Single Linkage Hierarchical Clustering

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2. Complete Linkage Clustering

Rule: distance between clusters = maximum distance between points.

Step-by-step merging:

1. Closest pair: P_2 and P_5 (distance = 0.02) $\Rightarrow$ merge to $\{P_2, P_5\}$
2. Update distances:

   $$d(\{P_2, P_5\}, P_1) = 0.90, \quad d(\{P_2, P_5\}, P_3) = 0.36, \quad d(\{P_2, P_5\}, P_4) = 0.53$$

3. Next closest: $\{P_2, P_5\}$ and P_3 (0.36) $\Rightarrow$ merge $\{P_2, P_3, P_5\}$
4. Update distances:

   $$d(\{P_2, P_3, P_5\}, P_1) = 0.90, \quad d(\{P_2, P_3, P_5\}, P_4) = 0.56$$

5. Next closest: $\{P_2, P_3, P_5\}$ and P_4 (0.56) $\Rightarrow$ merge $\{P_2, P_3, P_4, P_5\}$
6. Final merge: P_1 and $\{P_2, P_3, P_4, P_5\}$ (0.90) $\Rightarrow$ cluster $\{P_1, P_2, P_3, P_4, P_5\}$

Dendrogram: Complete Linkage

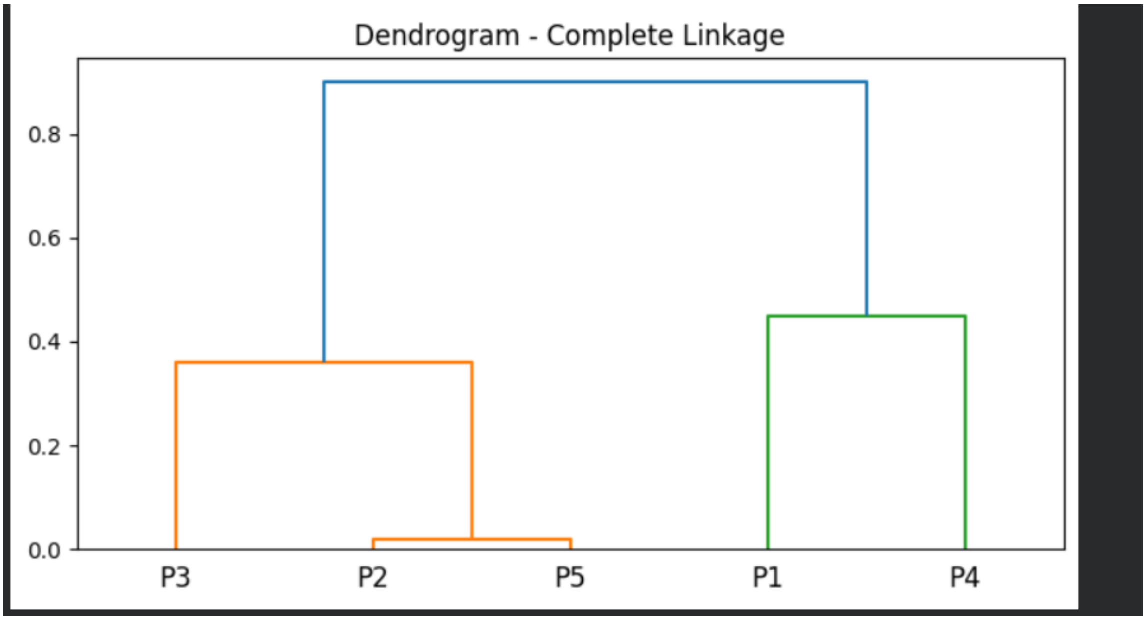

Figure 5: Dendrogram for Complete Linkage Hierarchical Clustering

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Summary

- **Single Linkage:** merges based on minimum distance → can produce "chained" clusters.
- **Complete Linkage:** merges based on maximum distance → produces compact, spherical clusters.
- Both dendrograms illustrate the order in which points/clusters are merged.

Hierarchical Clustering: Agglomerative vs Divisive

Hierarchical clustering builds a hierarchy of clusters. It has two main approaches:

Agglomerative (Bottom-Up)

- Start with each data point as its own cluster.
- At each step, merge the two closest clusters.
- Repeat until all points are in a single cluster.

Divisive (Top-Down)

- Start with all data points in a single cluster.
- At each step, split clusters into smaller clusters.
- Repeat until each cluster has only one point.

Comparison Table

Feature	**Agglomerative**	**Divisive**
Start	Each point as a single cluster	All points in one cluster
Approach	Merge clusters	Split clusters
Complexity	Less computationally intensive	More computationally expensive
Usage	More common	Less common
Dendrogram	Shows merging order	Shows splitting order